The Impact of Jet Fuel Prices And Passenger Demand On Airline Fares In The United States

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Introduction

The airline sector is very significant to the United States economic and transportation infrastructure. Air travel is the linking factor between cities, tourism, international trade and plays a key role in the growth of the economy and creation of employment. The air travel demand in the United States has grown significantly over the last decades as a result of increased income levels, globalization, population mobility, and growth of low-cost carriers. Nevertheless, airfare, which is also known as airline tickets prices, is not fixed and depends on numerous economic aspects. Two of the most influential determinants are passenger demand and jet fuel prices among others. One of the highest operating costs to airlines is jet fuel. Fuel costs in most instances represent a significant portion of total airline operating expenses, second to labor costs only. Owing to this, airlines pricing strategies are directly affected by changes in the prices of jet fuel. As fuel prices go up, the operating costs of airlines tend to rise and this can result in a rise of airfares by the airlines in a bid to stay afloat. On the other hand, airlines can be free to reduce ticket prices or increase their profit margins when fuel prices fall. The price elasticity of the world oil markets thus has a significant influence on the behavior of airline prices. Meanwhile, another factor that has a significant impact on airline fares is the demand among passengers. Like fundamental microeconomic laws, supply and demand play a major role in determining the price in a market. Demand to fly into and out of the country is higher, e.g., during the holiday seasons, economic booms or when consumer confidence is greater, and the airlines tend to charge more. Conversely, when demand is low due to economic recessions, external shocks or season, airlines can lower ticket prices to encourage demand and ensure that seats are occupied. As such, it is necessary to identify the trends in passenger demand when explaining changes in the airline fares. A case of study of these relationships can be interesting and relevant in the U.S. airline market. The structural changes that have affected the industry in the recent decades have comprised deregulation, merging of large carriers, emergence of low cost airlines, and growing competition. These have affected competitive strategies and pricing mechanisms in the market. Also, the airline sector is very vulnerable to external economic influences like the fluctuation of fuel prices, the macroeconomic cycles, and consumer behavior fluctuations. This has made airline fares to tend to adjust accordingly to the pressure on the cost-side as well as the demand changes. Microeconomically, the interplay of jet fuel prices, passenger demand and airline fare charges illustrates some basic economic concepts that include cost structures, price elasticity of demand and market equilibrium. Increased production cost in the form of higher fuel prices can shift the airline services supply curve upwards resulting in higher equilibrium prices in case of a constant demand. Equally, as the demand of the passengers rises, the demand curve will shift to the right, which may lead to an increase in the price of the tickets because the capacity of the

aircraft may not rise accordingly. The analysis of these relationships can be used to offer great information on how airlines react to the changes in costs and demand fluctuations to maximize their revenue and stay competitive. The other factor that is to be considered is the dynamic nature of the airline industry.

Airlines work in a very competitive and unpredictable environment in which the price choices have to be based on the operational expenses, the consumer demand and the strategic aspects. Dynamic pricing and revenue management systems are advanced pricing models that airlines often employ to change the fares according to the current demand pattern and cost situation. This complicates yet renders very pertinent to the economic analysis the relationship between fuel prices, passenger demand and ticket prices. The aim of this study is to investigate how jet fuel prices and passenger demand influence airline fares in the United States using a quantitative study that encompasses airfare prices, fuel prices, and passenger numbers throughout the years. Through the combination of these variables, the research is expected to determine whether fuel price hikes and passenger demand shifts play a major role in determining airfare. The data utilized in this study has monthly observations which can be used to examine trends and relations between these variables with time. It is the relationship that should be understood by a number of stakeholders. In the case of airlines, it assists in formulating effective pricing strategies and operational risks linked with the volatility of fuel prices. Policymakers and regulators can use the findings of such studies to assess the efficiency and competitiveness of the airline market. To the consumers, a better insight into the things that affect the change in airfare might give them a better view on the reason behind the price of tickets changing with time. This study is therefore aimed at adding to the already available literature on airline economics by examining the combined effect of cost-related variables (jet fuel prices) and demand-related variables (passenger numbers) on the airline fares in the United States. The study will determine whether these variables significantly influence the price of airfare and to a given extent, they will explain the price variations of airline tickets using a statistical analysis. The results of this research can help gain a clearer insight into pricing in the airline industry and suggest that both the costs of operations and the demand in the market can play a crucial role in shaping the level of airfare. Through an analysis of the relationship amongst jet fuel costs, passenger demand and airline fares, the study is an effective application of the microeconomic theory to a real world industry that is critical in the world economy.

2. Research Hypothesis

2.1 Overview & Theoretical Rationale :

This study investigates how two pivotal economic forces i.e. jet fuel prices and passenger demand shapes the airline fares in the United States. From price theory, the cost-push inflation framework, and demand-pull analysis, we can formulate the following hypotheses for empirical analysis.

Given the time-series nature of the data, the relationship between variables may differ in the short run and long run. Therefore, the analysis distinguishes between short-term dynamics and long-term equilibrium relationships.

Each hypothesis and it's economic theory:

H_1 - Uses cost-push inflation theory

H_2 - Uses demand-supply and yield management theory

H_3 - Ties both together

H_4 - Real world economic impact

2.2 The regression equation:

To capture both short-run and long-run dynamics, the model is estimated as follows -

1. Long-run (log-level model):

$$\mathbf{\ln(Airfare) = \beta_0 + \beta_1 \ln(Jet\ Fuel) + \beta_2 \ln(Passengers) + \varepsilon}$$

2. Short-run (difference-log model):

$$\mathbf{\Delta\ln(Airfare) = \beta_1 \Delta\ln(Jet\ Fuel) + \beta_2 \Delta\ln(Passengers) + \varepsilon}$$

(Where β_0 = intercept, β_1 & β_2 = regression coefficients,
ε = error term)

Dependent Variable : Airfare - average U.S. airline ticket price (USD)
Independent Variable 1 : Jet Fuel Price - cost per gallon/barrel of aviation fuel (USD)
Independent Variable 2 : Passengers - total domestic revenue passengers enplaned
Model Type : Log-level model (long run), Difference-log model (short run), Shock dummies (COVID), Interaction model

2.3 Hypothesis

1. Jet Fuel Prices -

H_{10} (Null): Jet fuel prices have no statistically significant effect on airline fares in the United States ($\beta_1 = 0$).

H_{11} (Alternative): Jet fuel prices have a significant positive effect on airline fares in the United States ($\beta_1 > 0$). (As jet fuel prices increase, airlines pass the higher operating costs onto consumers through elevated ticket prices.)

Fuel costs constitute approximately 20–30% of an airline's total operating expenditure (IATA, 2023). Under the cost-push theory of inflation, an exogenous rise in input costs such as crude oil price shocks compels firms to adjust output prices upward to preserve profit margins. Airlines, operating in a market with limited short-run flexibility to substitute fuel, are expected to translate rising fuel costs into higher fares, particularly on long-haul routes where fuel intensity is greatest.

2. Passenger Demand -

H_{20} (Null): Passenger demand has no statistically significant effect on airline fares in the United States
($\beta_2 = 0$).

H_{21} (Alternative): Passenger demand has a statistically significant effect on airline fares in the short run in United States. ($\beta_2 > 0$) (Higher passenger volume signals stronger market demand, enabling airlines to charge premium prices.)

According to standard demand-pull theory, increased demand holding capacity (supply) constant exerts upward pressure on equilibrium price. In the airline industry, this dynamic is amplified by yield management systems, whereby airlines dynamically price remaining seats higher as load factors rise. Periods of peak travel demand (holidays, summer travel) are historically associated with elevated average fares, consistent with a positive demand-price relationship.

3. Joint Effect -

H_{30} (Null): Jet fuel prices and passenger demand jointly have no significant explanatory power over airline fares
(all $\beta_i = 0$).

H_{31} (Alternative): Jet fuel prices and passenger demand, taken together, jointly and significantly explain the variation in airline fares in the United States. (The model will be evaluated using the F-test.)

Airline pricing depends on many factors. Fuel costs set the minimum price airlines need to cover their expenses, while passenger demand decides how high prices can go in the market. When both of these factors are included in the same regression model, it can explain airfare well because it captures both cost-related increases and demand-driven increases at the same time.

4. Economic Shocks -

H_{40} (Null): Major external shocks (such as COVID-19) don't have a significantly negative impact on airline fare levels.

H_{41} (Alternative): Major external shocks (such as COVID-19) have a significantly negative impact on airline fare levels.

2.4 Expected Signs & Directional Summary

- Jet Fuel Prices (β_1) → Positive (+) → Cost-push inflation; airlines pass higher input costs to consumers.
- Passengers (β_2) → Positive (+) → Demand-pull pricing; high demand raises equilibrium ticket prices.

2.5 Decision Rule

Each hypothesis will be evaluated using a two-tailed t-test at the 5% level of significance ($\alpha = 0.05$). The null hypothesis for each independent variable will be rejected if the computed p-value is less than 0.05, or equivalently if the absolute value of the t-statistic exceeds the critical value of approximately 1.96.

The overall model fit will be assessed using the F-test at the same significance level, alongside the coefficient of determination (R^2) to determine the proportion of variance in airfare explained by the two predictors.

3. Literature Review

The determination of airline fares has been extensively studied to understand industrial organization and price discrimination. Specifically, since deregulation under the Airline Deregulation Act of 1978, US airlines have had the freedom to set their own fares, making their pricing behavior a more pressing subject.

One of the variables which we will study that affect airline prices is jet fuel: one of the most significant and volatile cost components in airline operations. In 2014, jet fuel costs accounted for between 20% and 50% of total airline costs, underscoring their central role in the economics of aviation.(Atems et al., 2019)

However, after further analysis (Atems et al., 2019) also concluded that jet fuel prices cannot reliably predict the changes in airline prices.

On the other hand, (Dong et al., 2025) analyzed disparities across U.S. carrier types from 2000 to 2019, found that low-cost carriers are more inclined than legacy carriers to raise ticket prices in response to rising fuel costs, while ultra-low-cost carriers exhibit the lowest chance of changing costs. Specifically, when facing a 10% increase in fuel prices, legacy carriers' airfares rise by approximately 0.172%, while low-cost carriers' airfares increase by 0.351%.

According to the U.S. Energy Information Administration, jet fuel prices more than quadrupled in nominal terms between 2002 and 2013, from $0.72 per gallon to $2.98 per gallon. In response

to higher fuel prices, airline employment fell by approximately 23% between December 2002 and December 2013, from roughly 422,000 to 325,000 workers.(GAO, 2014)
According to (Sibdari et al., 2018) airlines also adjusted their capacity decisions in response to fuel price swings. Flight frequency and aircraft size move in opposite directions in response to changes in fuel costs and overall passenger demand, with fuel cost being a key factor in airlines' capacity decisions.

The responsiveness of air passengers to fare changes has been extensively studied, with the evidence strongly indicating that any policy action resulting in higher fares will produce a decline in demand. The magnitude of the decline, however, depends heavily on traveler type and route characteristics. Business travelers are generally less sensitive to fare changes than leisure travelers, a distinction that becomes particularly important during periods of demand shock. (IATA, 2007)
Contrary to this, (Richard, 2009) argued that the overall price elasticity of demand for domestic U.S. air travel has historically been inelastic, a finding that goes completely against general economic theory.
The type of airline business model(network carriers, low-cost carriers, charter carriers, and regional carriers) also monitors the relationship between ticket prices and demand. With (McConnachie et al., 2013) showing through their panel data analysis covering airlines from 2013 to 2022, that there is a significant and negative relationship between ticket price and airline demand, but also elasticity differences according to the business model.

One cannot examine US aviation history without including the September 11 attacks. In the immediate aftermath of the attacks, the U.S. government grounded the commercial fleet for three days, resulting in a 31.6% reduction in travel volume in September 2001 compared to the same month in 2000 and generating massive industry losses.(Clark et al., 2009) U.S. airlines lost $8 billion in 2001, and the industry was not profitable again until 2006, with cumulative losses topping $60 billion over that five-year period.(Josephs, 2021)
The structural changes triggered by the attacks reshaped the industry's competitive landscape. Low-cost carriers grew significantly in the aftermath of 9/11, air fares decreased, and network carriers responded to domestic market pressures by reducing available seats and shifting capacity to international routes, while dramatically cutting employment. Additionally, post-9/11 security policy itself became a demand suppressor. Another extreme shock was the Covid-19 pandemic. This produced a demand collapse far more severe than any prior crisis in commercial aviation history. According to the U.S. Department of Transportation, passenger traffic in April 2020 was 96% lower than April 2019, and remained 60% below 2019 levels throughout 2020.(GAO, 2021)

The pandemic's impact on fares was complex. Research analyzing airline pricing curves found an overall decrease in airline price per kilometer of approximately 31%, alongside reduced price

dispersion, as airlines adapted to a passenger mix increasingly dominated by price-sensitive leisure travelers and a general rise in travelers' price sensitivity.(Morlotti & Redondi, 2023) Airlines also restructured their pricing strategies in response. Industry discussions documented at the 2020 AGIFORS (Airline Group of the International Federation of Operational Research Societies) conference highlighted that the pandemic accelerated shifts toward dynamic and continuous pricing, with major carriers including Air Canada, United, and Qatar all indicating a greater focus on willingness-to-pay estimation and marginal revenue optimization. (Garrow & Lurkin, 2021)Comparisons between 9/11 and COVID-19 are interesting. As with 2001, companies were once again forced to restrict employee travel, recreating the collapse in premium business traffic that had devastated network carriers two decades earlier. However, unlike in 2001, the domestic market had already adapted to reduced premium revenues, meaning that long-haul international markets which were still highly dependent on premium fares faced the greater structural vulnerability during the pandemic.(Miller, 2020)

Overall, the literature establishes that jet fuel prices and passenger demand interact dynamically to shape airline fare levels, but that this relationship is moderated by carrier type, market structure and route characteristics. Exogenous disruptions, whether terrorism or pandemic only helps to reveal the structural fragility of demand-driven pricing models. However, there remains a gap in integrating these elements into a unified empirical framework, particularly in the context of the United States. Existing studies often examine fuel prices or demand separately, while fewer studies jointly estimate their impact on fares alongside major exogenous shocks. Gaps also remain in understanding the post-COVID calculation of fare structures and future research should also examine whether the traditional frameworks for understanding fare changes based on fuel and its demand elasticity remain valid in the post-pandemic pricing environment.

4. Data and Methodology

4.1 Data Description and Preliminary Processing

This study draws on a monthly dataset covering January 2000 through September 2025, giving 309 observations for the United States domestic airline industry. The three core variables are the air fare price index, jet fuel prices, and total passenger counts. All three variables were subsequently transformed into natural logarithms. This practice stabilises variance and reduces the leverage of extreme observations, and also means that the estimated coefficients can be read directly as elasticities. Two dummy variables were also constructed to account for well-known structural shocks. A dummy captures the four-month disruption following the September 11 attacks (September to December 2001), while another dummy marks the onset of the COVID-19 pandemic from March 2020 onwards.

The study has 3 core variables that are as follows:

1. Air fare (dependent variable): The data for this was acquired from U.S Bureau of Labor Statistics, an official U.S government database for U.S. Department of Labor.
2. Fuel prices (independent variable): data was acquired through FRED. This data is measured monthly. It is a proxy for fuel prices. It is defined as the Producer Price Index for jet fuel, which is used to capture input costs in the airline industry
3. Passengers: The dataset is collected from FRED and is also monthly time series data. It is described as revenue passenger miles, which is a measure for total volume of air travel. It accounts for the number of passengers, as well as for miles flown. It is a proxy for demand for airplanes.

4.2 Stationarity Testing

Before estimating any regression, it is important to establish whether the variables are stationary. Meaning, the fact that their statistical properties remain stable over time needs to be checked. Regressing non-stationary series on one another risks producing a spurious regression: one that appears highly significant but is driven entirely by shared trends rather than any genuine economic relationship.

The Augmented Dickey-Fuller (ADF) test was applied to each logged variable in levels, with both a constant and a deterministic trend included to reflect the upward trajectory visible in all three series. The results confirmed what a glance at the data already suggested: log air fare, log fuel price, and log passenger count are all non-stationary in levels. None of them reject the null hypothesis of a unit root at conventional significance levels. Once first-differenced, however, all three become stationary, with p-values of 0.01 or below.

```
        Augmented Dickey-Fuller Test

data:  data$dlfare
Dickey-Fuller = -8.2635, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary


        Augmented Dickey-Fuller Test

data:  data$dlfuel
Dickey-Fuller = -6.5634, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary


        Augmented Dickey-Fuller Test

data:  data$dlpassengers
Dickey-Fuller = -8.2292, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary
```

4.3 Multicollinearity Check

Before going ahead with the regression estimation, a Variance Inflation Factor (VIF) test was run to check whether fuel prices and passenger demand are too closely correlated with each other to be reliably separated in the regression. The VIF values for both independent variables came back close to 1, which means that there is no evidence of multicollinearity. Each variable's effect on air fares is being identified independently of the other.

```
> vif(model)
      dlfuel dlpassengers
    1.175168     1.175168
```

4.4 Model Specification

Given that all three variables are I(1), the model is estimated in first-differenced logs. The baseline specification is:

$$\Delta log(fare)_t = \beta_0 + \beta_1 \Delta log(fuel)_t + \beta_2 \Delta log(passengers)_t + \beta_3 D911_t + \beta_4 Dcovid_t + \varepsilon_t$$

where Δ denotes the first difference operator. Two additional models were estimated alongside this baseline: a log-level specification to examine long-run relationships, and an interaction model to test whether the fare-fuel and fare-passenger relationships changed structurally during the pandemic. The results from all three are as follows:

4.5 Diagnostic Tests

The test of serial autocorrelation performed by the Durbin-Watson method gave a statistic value of 1.21, with a probability of significance less than 0.001. It provides strong evidence of positive serial autocorrelation. The failure to correct for it would have resulted in underestimation of standard errors, implying more precise parameter estimation. In order to correct for this problem, heteroskedasticity and autocorrelation consistent Newey-West HAC standard errors were used. It is a known approach of inference, which remains valid in case of serial correlation.

According to the Breusch-Pagan test, the statistic was 1.16, and the p-value was 0.56. There is no indication of changing variance in the error terms; therefore, heteroskedasticity is not a problem in this particular model.

According to the Jarque-Bera test for normality of residuals, the statistic was 641.52, indicating a very small probability of a normally distributed error term. While at first sight, it is alarming, it actually poses no threat to estimation results. According to the Central Limit Theorem, an estimator of OLS becomes normally distributed in case of a large sample size, in spite of the distribution of errors.

```
> dwtest(model)

        Durbin-Watson test

data:  model
DW = 1.2258, p-value = 4.193e-12
alternative hypothesis: true autocorrelation is greater than 0

> bptest(model)

        studentized Breusch-Pagan test

data:  model
BP = 1.2564, df = 2, p-value = 0.5336

> jarque.bera.test(residuals(model))

        Jarque Bera Test

data:  residuals(model)
X-squared = 659.95, df = 2, p-value < 2.2e-16
```

4.6 Regression Results

Table 1 presents the results from the differenced log model

Table 1: Model 1 — Differenced Log Model

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	0.0008	0.0015	0.542	0.588
Δlog(Fuel Price)	0.0711	0.0154	4.612	0.000***
Δlog(Passengers)	0.0486	0.0088	5.549	0.000***
D911 (Sep–Dec 2001)	-0.0050	0.0079	-0.626	0.532
Dcovid (Mar 2020–)	-0.0072	0.0063	-1.131	0.259
R-squared	0.2227			
Adjusted R-squared	0.2124			
F-statistic	21.70	p-value	9.31e-16	***

The intercept of 0.0008 is statistically insignificant (p = 0.588). This is entirely expected in a first-differenced model — it simply means there is no autonomous monthly drift in fare growth beyond what fuel prices and passenger volumes can explain.

The coefficient on fuel price is 0.071, significant at the 0.1% level. In practical terms, a 1% rise in jet fuel prices is associated with a 0.07% increase in air fares within the same month, holding demand constant. The direction of this finding is exactly what economic theory would predict — airlines face higher costs and pass at least some of them on to passengers. What is perhaps more interesting is how small the elasticity is. Airlines do not fully and immediately pass through fuel cost increases. This is consistent with the widespread use of fuel hedging in the industry, as well as the competitive pressures that limit how quickly and how much any individual carrier can raise its prices.

0.0490 is the slope coefficient of the passenger demand which is equally significant at the 0.1 percent significance level. An additional 1% in passenger numbers leads to an increase in airfares by 0.05%. This follows economic theory on pricing in the sense that the more people travel, the greater the power to charge for tickets. Again, the elasticity is small, reflecting the fact that fare adjustments in response to demand shifts tend to be incremental rather than sharp — particularly in a competitive market with advance booking dynamics.

The September 11 dummy has a negative coefficient value of -0.0050 but is not statistically significant (p = 0.532). Although the fare rate did decline after the shocks, the short-run differencing model treats this effect as a large outlier. This is in line with the overall story – the shock to the airline industry was intense but temporary, and the industry returned to its original price path soon after.

However, the COVID-19 shock is another matter entirely. Not only was it negative (-0.0072), but like the September 11 dummy, the COVID-19 dummy also was not statistically significant (p = 0.259). The question to ask here is why. The reason it does not register as significant here is a modelling artefact: the first-difference specification captures month-on-month changes, and the pandemic showed up as extreme volatility in those changes — a catastrophic fall in early 2020 followed by a sharp rebound — rather than as a sustained directional shift in the growth rate of fares. This is precisely why the log-level model is also estimated.

4.7 Log-Level Model — Long-Run Relationships

Table 2: Model 2 — Log-Level Model (OLS)

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	4.6924	0.3190	14.711	< 2e-16***
log(Fuel Price)	0.1504	0.0113	13.260	< 2e-16***
log(Passengers)	0.0054	0.0187	0.289	0.773
Dcovid (Mar 2020–)	-0.1652	0.0289	-5.723	0.000***
D911 (Sep–Dec 2001)	0.0402	0.0274	1.465	0.144
R-squared	0.5494			
Adjusted R-squared	0.5435			
F-statistic	92.37	p-value:	< 2.2e-16	***

*Note: *** significant at 0.1%. OLS standard errors reported.*

The log-level model estimates the long-run relationship between fare levels, fuel prices, and passenger demand. In this specification, fuel prices retain a strong and significant effect: a 1% increase in fuel prices is associated with a 0.15% increase in fare levels — a notably larger elasticity than in the short-run model, suggesting that the long-run pass-through of fuel costs is more complete than the month-to-month adjustment would imply.

Passenger demand, however, is not statistically significant in the long-run model. This is an important and coherent finding: passenger volumes appear to influence short-term pricing dynamics — how fares move month to month — but do not drive the long-run level of fares.

Over the long run, it is fuel costs, rather than demand, that anchor the structural pricing level in the airline industry.

The COVID-19 dummy is highly significant in the log-level model, with fares declining by approximately 16.5% during the pandemic period. This confirms what was suspected from the differenced model: the pandemic's impact was primarily a level shock to air fares rather than a change in their short-run growth dynamics. The log-level model, which is better suited to capturing structural shifts of this kind, picks it up clearly.

4.8 Interaction Model — Did COVID Change Pricing Behaviour?

Table 3: Model 3 — Interaction Model (Differenced Log)

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	0.0007	0.0015	0.485	0.628
Δlog(Fuel Price)	0.0694	0.0163	4.258	0.000***
Dcovid	-0.0072	0.0064	-1.134	0.258
Δlog(Passengers)	0.0251	0.0571	0.439	0.661
Δlog(Fuel) × Dcovid	0.0136	0.0523	0.260	0.795
Δlog(Pass.) × Dcovid	0.0213	0.0588	0.362	0.717
R-squared	0.2223			
Adjusted R-squared	0.2094			
F-statistic	17.26	p-value:	5.03e-15	***

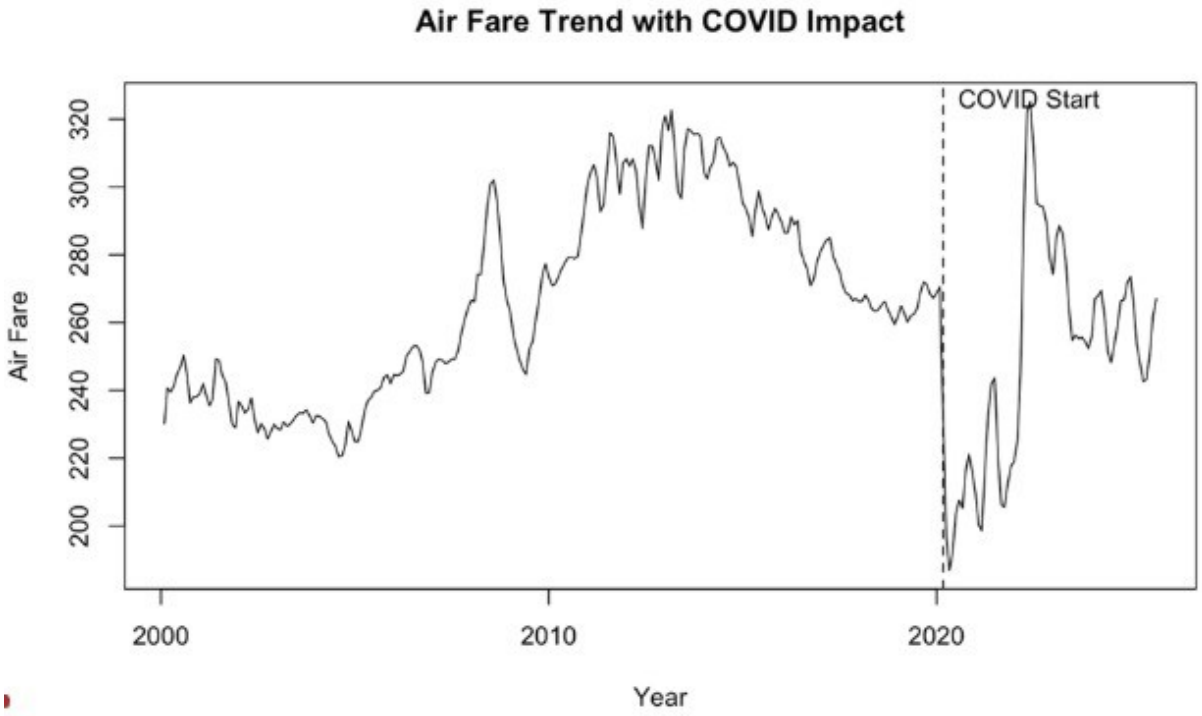
*Note: *** significant at 0.1%. OLS standard errors reported.*

A further specification was estimated to test whether the pandemic altered the fundamental relationship between fares and their determinants — in other words, whether airlines started responding differently to fuel price and demand signals during and after COVID. Interaction terms between the COVID dummy and both fuel price and passenger demand were included in the model.

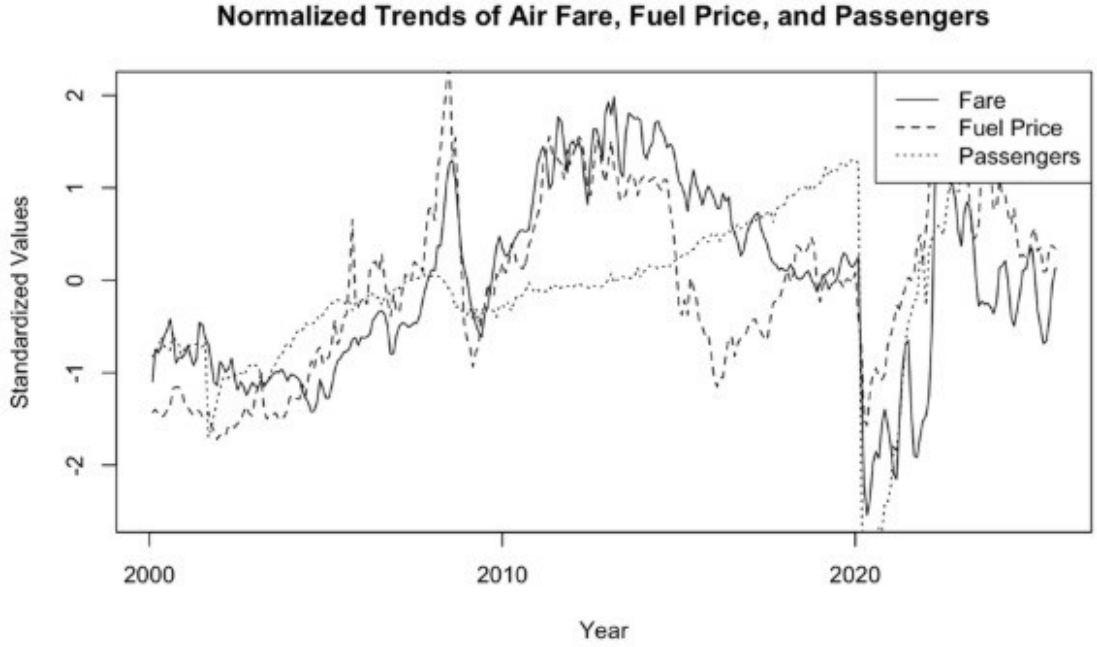
Neither interaction term is statistically significant. This is a meaningful result: it suggests that the pandemic did not structurally alter how airlines translate cost and demand signals into prices. The pricing mechanism itself remained intact. What had changed was the amount of fare charges, which were caused by reduced demand and disruptions in operations. The airlines still responded to prices of fuel and passenger numbers in more or less the same fashion, regardless of whether they were pre-pandemic or post-pandemic.

4.9 Visual Evidence

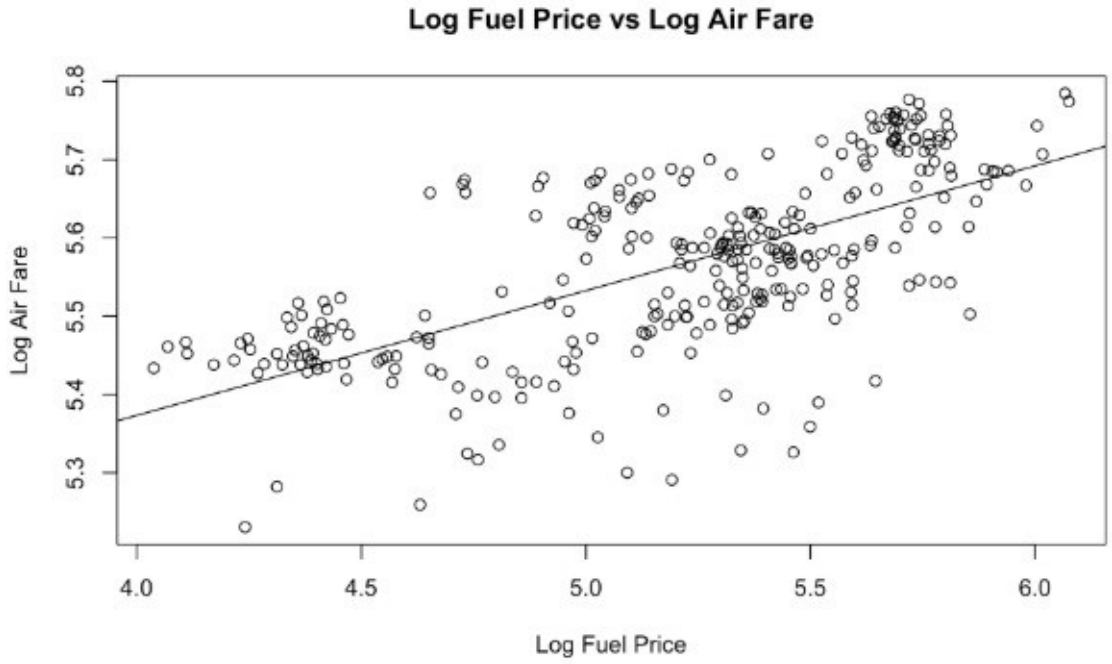
The graphical analysis of the data provides useful context for interpreting the regression results. The time series plot for air fares indicates that there was a period of stability from 2000 to about 2005. Thereafter, there were increasing levels of volatility. It should be noted that there is a sharp peak in 2014 after which the time series plot starts declining till 2020. It then takes a plunge towards its lowest level of fare. Afterward, the time series plot makes a V-shaped recovery to hit its highest-ever level in 2022.



A comparison of the time series plots for the three variables on a standardized basis shows that fuel prices and air fares had a strong correlation in 2008 and 2022 when the commodities experienced an upward shift. In the case of passengers, there was an uninterrupted growth rate from 2000 to 2019 and showed indifference to fare changes. The COVID outbreak proved to be a significant outlier in this case because of its simultaneous impact on the three variables.



The log-log scatter plot of fuel prices against air fares provides visual confirmation of the positive relationship estimated in the regression. The upward slope of the fitted line represents the fuel-fare elasticity directly. The scatter around the trend line is informative as well. While the correlation is clearly positive, the spread of observations illustrates that fuel is not the only driver of fares, and that other forces like demand conditions etc introduce meaningful variation around the trend.



In sum, the empirical findings paint a coherent picture. In the short term, fuel price increases and higher passenger demand both significantly influence the price of air tickets, with elasticities of 0.07 and 0.05 respectively. Both values are rather modest and indicate certain rigidity in pricing due to constraints faced by airlines. Fuel prices become the main determinant of fare levels in the long run, with much lesser significance of passenger demand. Shocks such as the novel coronavirus outbreak affect the level of the dependent variable and not its dynamic, which is confirmed by the interaction effect estimates. Once autocorrelation bias has been accounted for using Newey-West standard errors, all key results withstand scrutiny.

Conclusion

The analysis detailed here examines the microeconomic factors that influence airline ticket prices in the US. The primary focus of this research report is a look at how the price of jet fuel and passenger demand drive airline ticket prices. A comprehensive analysis over a long period of time; i.e. Year 2000 through Year 2025, has allowed researchers to develop empirical support for the basic economic principles regarding cost-push inflation and the relationship between supply and demand as it pertains to pricing strategy.

Based on this analysis, jet fuel prices are both a strong and statistically significant indicator of airline ticket prices for both short-term and long-term periods. In the short term, the researcher found that when fuel prices increased, ticket prices increased modestly; however, ticket price increases were subject to partial cost pass through as a result of competition and rigid factor market structures. Conversely, the long-term model illustrated that the elasticity of ticket price to an increase in fuel cost was greater than one, indicating that airlines will eventually make complete accommodations to the increase in cost of inputs by raising ticket prices to their new equilibrium price. This demonstrates that jet fuel prices are an important structural factor for establishing airline ticket pricing levels through time.

Passenger demand has a strong, but somewhat limited effect on pricing. Demand creates short-term increases in pricing and is supported by demand-pull theory and dynamic pricing; however, demand has no long run statistical correlation with fare level changes and therefore suggests that fare price fluctuations caused by demand are driven by changes to price in the short term.

The analysis also shows how an unexpected shock, such as COVID-19, is reflected in the short-run model as price volatility; however, when looking at longer-run models the decline in fare level during the pandemic is statistically significant. Additionally, according to the interaction analysis, these shocks did not change the way pricing is set under normal circumstances; they only impacted the level of pricing.

We can finally conclude that the price of airline tickets depends on two types of variables. Long-term trends in the prices that airlines charge are driven by cost factors (including the costs of fuel), in contrast to the longer-term price trends determined by cost variables (fuel prices), the short-run changes in the price of airline tickets depend on demand conditions. As a result, these findings continue to support both the importance of microeconomic theory as an explanation for real-world pricing behavior, and provide useful information for policy-makers, industry participants and consumers.

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